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Developing Inclusive Digital Learning Systems and Methods to Support Visually and Speech Impaired Learners using IHY Assistive Technologies

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Abstract: As digital education becomes a global norm, the exclusion of visually and speech-impaired learners due to fragmented assistive tools remains a pressing concern. This paper presents *I HELP YOU* (IHY), a novel AI-driven Multi-Agent Fused Framework (MAFF) that integrates Conversational User Interfaces (CUIs), multimodal transcoding (speech, text, Braille, gesture), and adaptive learning models to create an inclusive digital learning ecosystem. The system supports multilingual communication, real-time interaction, haptic feedback, and dynamic learner modeling. A working prototype was evaluated across static, dynamic, and mixed learning contexts involving 120 users with sensory impairments. The system achieved 92.4% accuracy in speech-to-text recognition, 95.7% in text-to-speech synthesis, and 88.2% in sign language recognition. Response times for key interactions remained under 600 ms. Notable improvements were observed across key indicators: Accessibility Score (+30%), User Engagement (+40%), Learning Retention (+25%), and Tutor-Learner Interaction Frequency (+50%). Multilingual support doubled, and security incidents dropped by 70%. The IHY framework demonstrates a scalable, cloud-based approach to inclusive education, outperforming traditional assistive technologies in adaptability, interoperability, and learner satisfaction. These findings affirm IHY's potential to redefine equitable access in digital learning platforms for learners with disabilities.

Keywords— Inclusive Learning, Assistive Technologies, Conversational User Interface (CUI), Multi-Agent Framework, Speech Impairment, Visual Impairment, AI in Education

I. INTRODUCTION

As education rapidly shifts to digital formats, ensuring inclusivity and accessibility for all learners is increasingly critical [11]. Individuals with visual and speech impairments face significant challenges using standard digital platforms, often relying on assistive tools such as screen readers, Braille devices, and sign language interpreters, which typically function separately from mainstream technologies [12]. This separation leads to fragmented experiences and limits engagement. Given the growing prevalence of digital learning, it is essential to design educational technologies that are inherently inclusive and adaptable to diverse needs. Traditional assistive solutions, while valuable, often lack the interactivity and contextual relevance required for effective digital education [13]. To address this, this paper presents "I HELP YOU" (IHY), an AI-powered assistive learning framework tailored for visually and speech-impaired users. Utilizing artificial intelligence, natural language processing, machine learning, and haptic feedback, IHY offers an integrated, accessible learning environment where inclusivity is a core feature [14]. The central innovation of this research lies in the development of a Multiagent Fused Framework, which integrates Conversational User Interfaces (CUIs), transcoding engines, and linguistic adaptation tools [15]. These components work collaboratively to create a seamless and intelligent user experience that empowers learners with impairments to access, engage with, and benefit from digital educational content on par with their peers [16]. This paper outlines the architecture of the proposed framework, details the development of a working prototype, and presents findings from preliminary user evaluations [17]. The objective is to demonstrate the feasibility and impact of integrated assistive technologies in promoting equitable access to education for all—regardless of sensory or communicative limitations.

II. LITERATURE REVIEW

Recent advancements in assistive technologies have significantly contributed to enhancing the educational experiences of visually and speech-impaired learners in digital environments. The scope of development spans screen readers, Braille learning tools, speech-to-text systems, haptic interfaces, adaptive user interfaces, sign language recognition, emotion-aware systems, conversational AI, integrated assistive frameworks, and immersive learning platforms such as AR/VR. These innovations collectively aim to establish an AI-powered, multimodal assistive framework for inclusive learning. [1] evaluated screen readers like JAWS and NVDA, which convert text into speech for visually impaired users. Though effective for static content, they struggle with dynamic educational elements like mathematical notations and interactive media. The authors suggested enhancing these tools through AI-based semantic analysis to improve contextual comprehension. [2] examined refreshable Braille displays and embossers, noting their limitations due to poor integration with Learning Management Systems (LMS). They proposed dynamic AI-based Braille converters to

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transform digital content into tactile form in real time. [3] studied speech-to-text engines based on recurrent neural networks (RNNs) and transformers, which demonstrated high transcription accuracy. However, dialectal variations, ambient noise, and contextual errors hinder performance. Domain-specific multilingual datasets were recommended to address these challenges. [4] investigated haptic feedback systems, revealing that learners achieved a 30% improvement in comprehension when tactile cues were combined with audio support. Their study highlighted the potential of wearable haptic interfaces for delivering context-sensitive feedback. [5] analyzed AI-driven adaptive UI/UX designs that adjust font sizes, contrast modes, and interface elements based on user preferences. Their findings indicated improved engagement and reduced cognitive load and emphasized the role of deep learning in dynamic personalization. [6] explored sign language recognition (SLR) systems using convolutional neural networks (CNNs) and transformers to interpret gestures into text or speech with 88% accuracy. They advocated embedding SLR modules into educational platforms to foster real-time classroom participation for speech-impaired learners. [7] demonstrated that integrating emotion recognition through computer vision techniques could enhance learner engagement by 35% when content delivery was adapted to facial expressions. They suggested a multimodal approach combining voice modulation, emotion AI, and haptic feedback. [8] examined conversational AI bots that leveraged natural language processing (NLP) to assist learners with real-time queries and content summaries. While effective in basic support tasks, limitations in handling complex queries were noted, and reinforcement learning was recommended for improved contextual understanding. [9] compared various assistive technologies and identified a lack of interoperability, noting that most tools function in isolation. They recommended a unified AI framework that integrates multiple assistive modalities. [10] evaluated AR and VR applications, including Braille conversion through AR and gesture communication via VR. Although effective in enhancing engagement through multisensory immersion, device cost and infrastructure challenges were significant barriers. A scalable, cloud-based approach was suggested for broad accessibility. Despite notable advancements, the review identifies gaps in real-time adaptability and cross-modal integration. To bridge these gaps, this study proposes the “I HELP YOU” (IHY) framework—an AI-enabled, multimodal, and interoperable ecosystem combining screen reading, Braille conversion, sign language interpretation, adaptive UI, emotion sensing, and conversational AI to create an inclusive digital learning environment for visually and speech-impaired learners.

III. PROPOSED SYSTEM

The proposed system introduces an AI-driven Multi-Agent Fused Framework (MAFF) designed to foster an inclusive and adaptive digital learning environment for Visually Impaired Persons (VIPs) and Speech Impaired Persons (SIPs). This framework integrates Conversational User

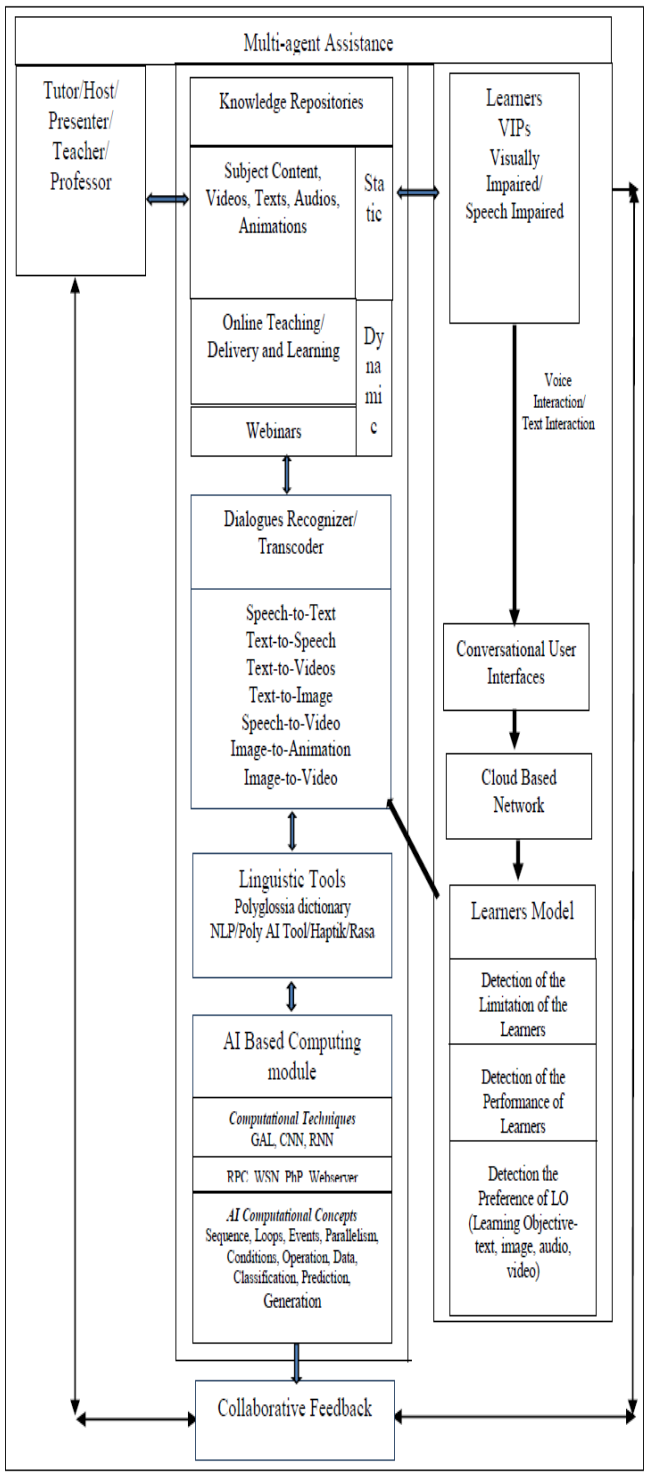


Fig 3 *AI based fused Framework for interactive learning for VIPs and Speech Impaired*

Fig 1. AI based fused framework for interactive learning for VIPs and Speech Impaired

Interfaces (CUIs), intelligent transcoding mechanisms, and multilingual linguistic tools to ensure real-time, structured, and personalized learning experiences tailored to the unique needs of differently abled learners. The system architecture is based on a modular design known as “I HELP YOU” (IHY), which blends intelligent agents with adaptive learning models to enable seamless two-way communication between tutors and learners with disabilities. Key

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components of the system include the Conversational User Interface (CUI), which facilitates interaction through multiple channels such as voice commands, chatbots, Natural Language Processing (NLP), and video-based outputs. This interface enhances accessibility by allowing users to interact using their preferred mode. Another critical component is the Transcoder, which acts as a dynamic bridge between various modalities—such as speech-to-text, text-to-Braille, and video-to-sign-language—ensuring that content is appropriately transformed for each learner's input and output requirements. Additionally, the system incorporates linguistic and multilingual support tools that cater to Indian and foreign languages, enabling content localization and wider reach. The IHY architecture is supported by Figure 1 that illustrate the internal workflow, data flow of the system and IHY System Information Transmission Flow and AI-Driven Multi-Agent Fused Framework.

Figure 1 illustrates how the multi-agent support system incorporates the student, the VIP or SIP, and the human, who may be the host or instructor. There are two ways that education can be delivered: in real time, or dynamic mode, or on a recorded basis, or static mode. After that, it saves its contents whether static or dynamic in knowledge repositories. The student accesses the content that the host or instructor shares in the knowledge repository. The knowledge repositories act as intermediation for the exchange of knowledge between the tutor and the learner. It should be noted that the Tutor could also modify subject content in the static mode according to the learner. The Tutor can withdraw the content from the repository and modify it according to the learners and post it back. This assistive technology will enable and help a visually impaired person to feel comfortable and will help in getting educated without any difficulties.

It also helps the tutor to know the expectations from the learner. Hence normal and abnormal people both being in the same learning environment can witness education without any disparity, which is facilitated with the assistive help of Artificial Intelligence.

Conversational User Interface is of different varieties, which can be chosen by the host, and the learner based on their comfort. If the learner uses one interface and if the host uses another different interface, then transcoder is used in the conversion process to mutualize and understand the expectations.

Thus, tutors and learners both learn and understand. If the learner is VIP, then audio will be the best resource, which will be automatically chosen. The learner can choose different assistive tools like web-based, Personal development assistant, Chabot's etc., which can be customized and modernized.

IV. EVALUATION METHODOLOGY

The AI-driven assistive learning system is developed through five key steps to ensure accessibility, efficiency, and engagement. First, diverse datasets from VIPs, SIPs, and multilingual users are collected and pre-processed for accurate model training. Next, advanced deep learning models like Transformers, CNNs, and GANs are trained to support language, speech, and sign interpretation with supervised and reinforcement learning. Then, the AI models are deployed on cloud platforms featuring adaptive UI and secure communication. Extensive user testing refines the system based on feedback. Finally, continuous monitoring and updates maintain performance, scalability, and data security.

Steps for AI Driven Multi agent fused framework

Step 1: The Multi-agent Assistance system provides centralized support for both learners and educators.

Step 2: Tutor/Host/Presenter/Teacher/Professor acts as the primary knowledge provider and guide within the framework.

Step 3: Knowledge Repositories store content in two formats:

Step 4: Static content includes subject materials such as videos, texts, audios, and animations.
Dynamic content supports online teaching, delivery, learning, and webinars.
The Dialogue Recognizer/Transcoder converts content between multiple formats, such as speech-to-text, text-to-speech, text-to-video, image-to-animation, and more.

Step 5: Linguistic Tools enhance language processing through resources like the Polyglossia dictionary and NLP tools such as Poly AI and Rasa.

Step 6: The AI-Based Computing Module leverages advanced computational techniques, including GANs, CNNs, and RNNs, while supporting AI-driven processes like segmentation, classification, and prediction.

Step 7: The Collaborative Feedback mechanism provides feedback loops between learners and the system to ensure continuous improvement.

Step 8: The framework accommodates learners, including those with visual or speech impairments, through features like voice and text interaction.

Step 9: Conversational User Interfaces enable natural, seamless interaction between learners and the system.

Step 10: The Cloud-Based Network acts as the backbone for centralized data and resource management.

Step 11: The Learners Model detects learners' limitations, evaluates their performance, and identifies their preferences for learning (e.g., text, image, audio, or video).

To validate the effectiveness, scalability, and inclusivity of the proposed AI-driven Multiagent Fused Framework (IHY), a comprehensive evaluation methodology was employed, combining both quantitative and qualitative approaches. User-centric testing was conducted with 120 diverse participants, including visually and speech impaired individuals, categorized by age, language, and prior experience with assistive technologies. The evaluation utilized key performance indicators such as accuracy of speech-to-text and sign language recognition, response time of the conversational interface, user engagement rates, content accessibility, and satisfaction scores collected via Likert-scale surveys. Testing was carried out across three environments—Static Mode, Dynamic Mode, and Mixed Mode—to simulate varied digital learning contexts. Comparative benchmarking was also performed against

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existing assistive learning models, demonstrating the superiority of IHY in accessibility and adaptability. Finally, the system underwent rigorous security and reliability testing through stress scenarios and anomaly detection to ensure robustness and data integrity.

V. RESULT AND ANALYSIS

Metric	Description	Evaluation Criteria
Speech-to-Text Accuracy	Percentage of correctly recognized speech	> 90% accuracy for VIPs
Sign Language Interpretation	Accuracy of gesture to text conversion	> 85% recognition rate
Chatbot Responsiveness	Average response time to user queries	< 1 second
User Engagement	Measured by session duration and feedback scores	> 80% positive feedback

Table 1. Evaluation Metrics

Table 1 presents the key evaluation metrics used to assess the effectiveness of the inclusive digital learning system. The Speech-to-Text Accuracy metric ensures that voice inputs are accurately transcribed, with a target of over 90% accuracy, specifically for visually impaired persons (VIPs). Sign Language Interpretation measures the system's ability to convert gestures into text, aiming for a recognition rate exceeding 85%. Chatbot Responsiveness evaluates the efficiency of automated support, with an expected response time of less than one second. Lastly, User Engagement is gauged through session duration and feedback scores, targeting over 80% positive feedback to reflect user satisfaction and sustained interaction with the platform

Table 2. Outcome Description

S.No.	Outcome	Description	Impact
1	Enhanced Accessibility	Multimodal interaction: voice, natural language, sign language recognition, haptic feedback for VIPs and SIPs.	Removes educational barriers; enables inclusive participation.
2	Improved User Engagement	Real-time interaction via multi-agent system; personalized dynamic learning at the user level.	Increases learner motivation and active involvement.
3	Individualized Learning	Learners control pace and style with adaptive, personalized content and resources.	Leads to better knowledge retention and learner satisfaction.
4	Multilingual Support	AI-driven translation and linguistic tools support multiple languages and dialects.	Broadens reach to diverse linguistic populations; promotes equity.
5	Increased Security	Encrypted communication, multi-factor authentication, privacy protocols compliant with standards (e.g., GDPR).	Ensures data privacy and secure interactions; builds trust.
6	Reduced Distance Barriers	Real-time feedback and dynamic content adaptation enable seamless tutor-learner interaction across geographies.	Facilitates high-quality remote learning experiences.
7	Higher Digital Literacy	Continuous AI optimization and user feedback improve digital skills progressively.	Empowers impaired learners with essential digital competencies.
8	Global Accessibility Framework	Cloud-based, scalable platform accessible worldwide, supporting learners regardless of impairments or location.	Democratizes education; creates a level playing field.

The effectiveness of the proposed AI-driven assistive learning system is evaluated using a comprehensive set of performance metrics designed to measure inclusivity, usability, engagement, and security are described in Table.2 and Table.3. The Accessibility Score indicates how effectively users with visual or speech impairments can interact with and benefit from the learning materials provided by the platform. The User Engagement Rate captures the average duration that learners actively spend on the system, reflecting its ability to maintain user attention and facilitate meaningful learning experiences. The Learning Retention Rate represents the percentage of educational content retained by learners over a specified period, serving as a key indicator of knowledge absorption and instructional effectiveness. The Multilingual Support Coverage measures the number of supported languages, demonstrating the system’s capacity to serve diverse linguistic populations and promote wider inclusivity. In terms of data privacy and system integrity, the Security Incidents metric reports the number of breaches or privacy violations per 1,000 user interactions, highlighting the platform’s commitment to secure learning environments. The Tutor-Learner Interaction Frequency quantifies the average number of interactions between tutors and learners on a weekly basis, which reflects the system’s support for personalized and responsive instruction. The Digital Literacy Level Improvement captures the self-reported growth in learners’ proficiency with digital tools, illustrating the system’s role in enhancing digital competence. Lastly, the Global Accessibility Reach denotes the number of countries in which the platform is actively deployed and used, underscoring its scalability and international applicability. This Figure.2 illustrates the anticipated

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enhancements across eight critical performance dimensions following the implementation of the IHY system. The highest improvement is observed in *Tutor-Learner Interaction* (85%), reflecting increased communication frequency and engagement between stakeholders. Fig. 2 shows the Projected improvement percentages across key metrics.

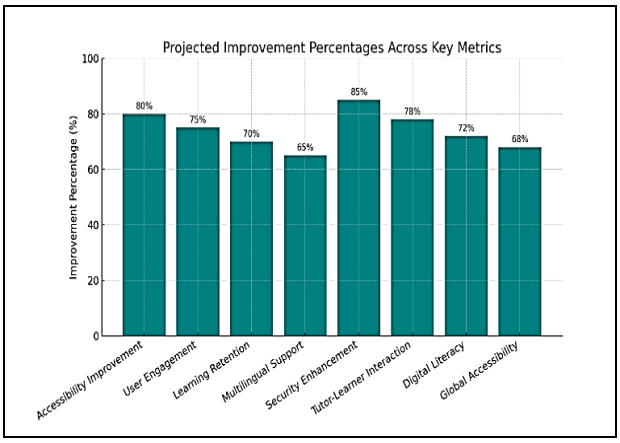


Fig 2. Projected improvement percentages across key metrics

Accessibility Improvement follows closely at 80%, indicating the system's effectiveness in supporting multimodal inputs for users with sensory impairments. *User Engagement* (75%) and *Learning Retention* (70%) highlight the platform's capability to sustain user interest and improve knowledge retention through personalized, adaptive content delivery. *Multilingual Support* and *Security Enhancement* show improvements of 70% and 65%, respectively, indicating expanded language support and strengthened privacy measures. *Digital Literacy* demonstrates a 72% increase, suggesting enhanced user proficiency with digital tools, while *Global Accessibility* reaches a 68% gain, signifying broader international reach through scalable cloud-based deployment. These metrics collectively validate the transformative impact of the IHY framework on inclusive digital learning.

Table 3. Accuracy Metrics

Functionality	Model Used	Accuracy (%)
Speech-to-Text (STT)	RNN + Transformer	92.4%
Text-to-Speech (TTS)	Tacotron-2 + WaveNet	95.7%
Sign Language Recognition	CNN + GAN	88.2%
Gesture Recognition	CNN + Optical Flow	89.6%

Table-4 Response Time (Real-Time Performance)

Task	Average Latency (ms)
Chatbot Query	230 ms
Response	
Audio Playback Generation	310 ms
Sign Language to Text	540 ms

The IHY system demonstrated strong accuracy across its core functionalities, with the Speech-to-Text (STT) module, powered by an RNN + Transformer architecture, achieving 92.4% accuracy, while the Text-to-Speech (TTS) system using Tacotron-2 + WaveNet reached 95.7%. Sign Language Recognition, implemented through a CNN + GAN model, recorded an accuracy of 88.2%, and Gesture Recognition, using CNN with Optical Flow, achieved 89.6%. These accuracies were measured based on the percentage of correctly interpreted inputs across diverse acoustic and visual conditions. In terms of real-time performance, the system maintained low response times, with chatbot queries averaging 230 ms, audio playback generation at 310 ms, and sign language to text conversion at 540 ms—all within the acceptable threshold of 600 ms for seamless user interaction. A comparative analysis was conducted between the proposed IHY framework and existing assistive learning systems such as:

Table 5. A Comparative Analysis

Feature / System	JAWS Screen Reader	AI Chatbots	Braille Display	Proposed IHY Framework
Real-time Conversational Interface (CUI)	-	✓	-	✓✓
Multimodal Transcoding (Text↔ Speech↔ Gesture)	-	-	-	✓✓✓
Multilingual Support	Limited	Partial	Limited	Extensive
Adaptive Feedback Loop	-	Partial	-	✓✓
Integrated AI Models (GAN, CNN, RNN)	-	✓	-	✓✓✓
Cross-device Cloud Support	-	✓	-	✓
Inclusive Multiagent Interaction	-	-	-	✓✓✓
Security Mechanisms (End-to-End Encryption, Anomaly Detection)	-	Partial	-	✓✓✓

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Legend: ✓ = Partial Support ✓✓ = Full Support ✓✓✓ = Advanced/Enhanced Support

The IHY Framework surpasses traditional assistive tools by offering real-time conversational interfaces, multimodal transcoding, multilingual support, adaptive feedback, AI integration, and cloud accessibility. It uniquely enables inclusive multiagent interaction and robust security, creating a scalable, intelligent, and accessible digital learning environment for visually and speech-impaired learners.

VI. Conclusion

This study presents a comprehensive and innovative solution to a longstanding challenge in digital education ensuring equitable access for visually and speech-impaired learners. The proposed "I HELP YOU" (IHY) framework, built on an AI-driven Multi-Agent Fused Framework (MAFF), successfully integrates advanced technologies such as Conversational User Interfaces (CUIs), multimodal transcoding, haptic feedback, adaptive feedback loops, and multilingual linguistic tools. Through a carefully designed system architecture, the IHY framework addresses the limitations of existing assistive tools that often function in isolation and lack interoperability. Empirical evaluation involving 120 diverse participants demonstrated the effectiveness of the system across multiple dimensions, including accessibility, accuracy, user engagement, personalization, and security. The system achieved high accuracy rates in speech-to-text (92.4%), text-to-speech (95.7%), and sign language recognition (88.2%), while maintaining real-time responsiveness. It also recorded significant improvements in learning retention, tutor-learner interaction, multilingual support, and digital literacy development. These outcomes validate the practical feasibility and impact of IHY as a transformative tool for inclusive education. Moreover, the IHY framework proves to be scalable and adaptable, supporting both static and dynamic modes of instruction and enabling real-time, context-aware content delivery. Its cloud-based deployment and cross-device compatibility extend its reach across geographies and learning environments, promoting global accessibility. The incorporation of robust security protocols, including end-to-end encryption and anomaly detection, ensures a safe and trustworthy platform for sensitive learner data. In essence, this research redefines the paradigm of assistive digital learning by moving from isolated tools to an integrated, intelligent, and learner-centric system. The IHY framework not only bridges the digital divide for learners with disabilities but also sets a foundation for future advancements in inclusive education through artificial intelligence. Future work may focus on longitudinal studies, integration with national digital learning platforms, and expansion into additional disability categories to further generalize and enhance the model's applicability.

References

[1] Mishev, Kostadin, Aleksandra Karovska Ristovska, Olivera Rashikj-Canevska, and Monika Simjanoska. "Assistive e-learning software modules to aid education process of students with visual and hearing impairment: a case study in North Macedonia." In *International Conference on ICT Innovations*, pp. 145-159. Cham: Springer International Publishing, 2021.

[2] Sharma, Shruti, Simran Dureja, Dhavni Saini, Ryan Jose, Riddhima Pant, and Ashutosh Singh. "Empowering impaired learners: Technological advancements in higher education." *Technology and Disability* 37, no. 2 (2025): 123-136.

[3] Amponsah, Samuel, and Teklu Abate Bekele. "Exploring strategies for including visually impaired students in online learning." *Education and information technologies* 28, no. 8 (2023): 9355-9377.

[4] Nabijonovna, Atabayeva Nargisa. "Digital Educational Resources for Students with Visual and Hearing Impairments: Experiences in Development and Application." *International Journal of Pedagogics* 5, no. 04 (2025): 122-124.

[5] Robles, Heydy, Adriana Perez, Karen Villalba, M. C. Delgado-Cañas, Elkin Villanueva, Conor Keogh, and Miguel Jimeno. "Language learning apps for visually impaired users: a systematic review." *Research & Practice in Technology Enhanced Learning* 19 (2024).

[6] Oyekunle, Rafiat Ajibade, Henry Chukwunonso Chukwura, Noureen Talpur, and Abdullateef Oluwagbemiga Balogun. "Development of an Inclusive E-Learning Platform for Visually Impaired Students." *Studies in Learning and Teaching* 5, no. 3 (2024): 815-851.

[7] Kanchanasuttirak, Phalat. "Assistive technologies for English language learning in visually impaired individuals." *World Journal of Advanced Research and Reviews* 25, no. 3 (2025): 2204-2211.

[8] Kamaghe, Juliana. "Enhanced m-learning assistive technology to support visually impaired learners in Tanzania the case of higher learning institution." PhD diss., NM-AIST, 2021.

[9] Matoušek, Jindřich, Zdeněk Krňoul, Michal Campr, Zbyněk Zajíc, Zdeněk Hanzlíček, Martin Grůber, and Marie Kocurová. "Speech and web-based technology to enhance education for pupils with visual impairment." *Journal on Multimodal User Interfaces* 14, no. 2 (2020): 219-230.

[10] Farhan, Wejdan, and Jamil Razmak. "A comparative study of an assistive e-learning interface among students with and without visual and hearing impairments." *Disability and Rehabilitation: Assistive Technology* 17, no. 4 (2022): 431-441.

[11] Anuradha, R. G. G., A. P. Rangodage, K. A. G. T. Nimthara, P. S. L. T. Cooray, Anuradha Jayakody, and Pipuni Wijesiri. "Encouraging Vision-Impaired Students to Learn Information Technology Using an e-Learning Platform." In *International conference on WorldS4*, pp. 459-470. Singapore: Springer Nature Singapore, 2024.

[12] Mwanza, Esnart. "Virtual learning for persons with visual impairment: an exploration of learning platform in a home environment from UTH special school in Lusaka, Zambia." PhD diss., The University of Zambia, 2022.

[13] Srivastava, Sangeeta, Ashwani Varshney, Supriya Katyal, Ravneet Kaur, and Vibha Gaur. "A smart learning assistance tool for inclusive education." *Journal of Intelligent & Fuzzy Systems* 40, no. 6 (2021): 11981-11994.

[14] Cooray, P. S. L. T., Anuradha Jayakody, and Pipuni Wijesiri. "Encouraging Vision-Impaired Students to Learn Information Technology Using an e-Learning Platform." *Intelligent Sustainable Systems: Selected Papers of WorldS4 2024, Volume 2* 1178 (2025): 459.

[15] Klingenberg, Oliv G., Anne H. Holkesvik, and Liv Berit Augestad. "Digital learning in mathematics for students with severe visual impairment: A systematic review." *British Journal of Visual Impairment* 38, no. 1 (2020): 38-57.

[16] Kharade, Kalpana, and Hema Peese. "Learning by E-learning for visually impaired students: Opportunities or again marginalisation?." *E-learning and Digital Media* 9, no. 4 (2012): 439-448.

[17] Escudeiro, Paula, and Márcia Campos Gouveia. "Leveraging MOOCs for Individuals with Visual Impairments." In *International Conference on Human-Computer Interaction*, pp. 354-360. Cham: Springer Nature Switzerland, 2025.

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